RFM Customer Segmentation and Customer Retention Analysis

1. Project Overview

Project Title: RFM Customer Segmentation and Customer Retention Analysis
Domain: E-commerce / Retail Analytics
Dataset: UCI Online Retail II
Tools Used: Python, Pandas, NumPy, Jupyter Notebook, Power BI

Business Problem

E-commerce businesses need to understand customer purchasing behaviour to improve customer retention, identify valuable customers, and use marketing budgets effectively.

Treating every customer in the same way can lead to missed retention opportunities. For example, a customer who purchases frequently and spends a large amount should not receive the same marketing treatment as a customer who has not purchased for a long time.

The objective of this project was to segment customers based on their historical purchasing behaviour using the RFM framework and identify actionable opportunities for customer retention and reactivation.

Project Objectives

- Analyze historical e-commerce transactions.
- Clean and prepare the data for customer-level analysis.
- Calculate Recency, Frequency, and Monetary values for each customer.
- Assign RFM scores and classify customers into meaningful segments.
- Identify high-value customers and customers who may be at risk of becoming inactive.
- Develop targeted marketing recommendations.
- Present the findings through a Power BI dashboard.

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2. Dataset Description

The project uses the Online Retail II dataset from the UCI Machine Learning Repository. It contains transactional records from an online retail business.

The dataset includes two annual transaction files covering the periods 2009–2010 and 2010–2011.

Important Columns

Column	Description
Invoice	Unique invoice or transaction identifier
StockCode	Product identifier
Description	Product description
Quantity	Number of units purchased
InvoiceDate	Date and time of the transaction
Price	Unit price
Customer ID	Customer identifier
Country	Customer's country

A sales value was calculated from quantity and unit price to support revenue and monetary analysis.

Data Preparation Challenges

The dataset contained missing customer identifiers, duplicate records, and transactions that required validation before customer-level analysis could be performed.

Because RFM analysis depends on identifying individual customers, records without a Customer ID could not be reliably assigned to a customer segment.

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3. Data Cleaning and Preparation

Data cleaning was performed in Python using Pandas and NumPy.

The main steps were:

1. **Combined the datasets:** Loaded the two annual datasets and combined them into a single dataset for analysis.
2. **Reviewed missing values:** Checked missing values in important columns, especially Customer ID and Description.
3. **Checked duplicate records:** Identified duplicate transactions during data-quality assessment.

4. **Handled missing Customer IDs:** Excluded transactions without a usable Customer ID from customer-level RFM analysis.
5. **Validated quantities and prices:** Removed invalid negative-quantity records and filtered out zero or negative prices according to the notebook's cleaning rules.
6. **Excluded non-product transaction entries:** Removed specified entries such as postage, carriage, manual adjustments, and bank charges from the product-sales analysis.
7. **Created sales values:** Calculated sales as Quantity × Price.
8. **Prepared transaction dates:** Converted InvoiceDate into a suitable date-time format and used the latest transaction date to establish the analysis reference date.

The reference date for the revised analysis was **10 December 2011**.

Why Data Cleaning Was Important

Cleaning helped reduce the impact of invalid transactions and non-product entries on customer metrics. It also ensured that the RFM calculations were based on identifiable customers and usable purchase records.

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4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand the transaction data before calculating customer segments.

The analysis focused on:

- Overall sales revenue.
- Number of orders and products.
- Number of unique customers.
- Average order value.
- Customer purchase frequency.
- Customer spending behaviour.
- Customer inactivity and potential retention opportunities.

Key Business Metrics

The revised notebook produced the following results:

Metric	Revised result
Total revenue	£17,374,804.27
Total orders	36,975
Unique customers in RFM analysis	5,881
Unique products	4,631
Average order value	£469.91
Average Recency	200.99 days
Average Frequency	6.29 orders
Average Monetary value	£2,954.40

Revenue and monetary values are expressed in the dataset's currency, GBP.

Average Order Value

Average Order Value (AOV) measures the average revenue generated per order.

Formula:

AOV = Total Revenue ÷ Total Orders

Using the revised results:

£17,374,804.27 ÷ 36,975 ≈ £469.91

This metric provides a useful baseline for understanding the value of an average order.

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5. RFM Methodology

RFM stands for Recency, Frequency, and Monetary value. It is a customer segmentation method that uses historical purchase behaviour to help businesses understand customer value and engagement.

5.1 Recency

Recency measures how recently a customer made a purchase.

Formula:

Recency = Reference Date − Date of Most Recent Purchase

A lower Recency value indicates a more recent purchase and generally stronger recent engagement.

5.2 Frequency

Frequency measures how often a customer purchases.

In this project, Frequency was calculated using the number of unique invoices associated with each customer.

A higher Frequency value indicates that the customer has placed more orders during the available observation period.

5.3 Monetary Value

Monetary value measures the total sales value generated by a customer.

Formula:

Monetary Value = Sum of Sales for Each Customer

A higher Monetary value indicates greater historical spending.

5.4 RFM Table

The three metrics were aggregated at customer level to create an RFM table containing one row per customer.

The revised RFM table contains **5,881 customers**.

Metric	Average value
Recency	200.99 days
Frequency	6.29 orders
Monetary	£2,954.40

These averages provide an overall view of customer purchasing behaviour. They do not, by themselves, describe the behaviour of every individual customer.

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6. RFM Scoring

After calculating the three RFM metrics, customers were assigned scores using five-point quantile-based scoring.

Each customer received:

- R score for Recency.
- F score for Frequency.
- M score for Monetary value.

Each component uses a score from 1 to 5.

Scoring Logic

Recency: Customers who purchased more recently receive higher scores.

Frequency: Customers who place more orders receive higher scores.

Monetary: Customers who spend more receive higher scores.

Recency is scored in the reverse direction because a lower number of days since the last purchase is more desirable. Frequency and Monetary scores follow the normal direction, with higher values receiving higher scores.

Overall RFM Score

The overall RFM score was calculated by adding the three component scores.

Formula:

Overall RFM Score = R Score + F Score + M Score

The theoretical range is 3 to 15.

A higher score generally indicates stronger customer purchasing behaviour across the three dimensions. However, the overall score should be interpreted alongside the individual R, F, and M values and the assigned segment.

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7. Customer Segmentation

Customers were grouped into eight segments using the RFM scores and the segmentation rules implemented in the revised notebook.

The objective was to translate numerical scores into meaningful customer groups that can support business decisions.

Revised Customer Segment Distribution

Customer segment	Customers
Champions	1,291
Hibernating	1,280
Loyal Customers	1,147
Needs Attention	1,134
At Risk	389

Potential Loyalists	359
Can't Lose Them	224
New Customers	57
Total	**5,881**

7.1 Champions

Customers: 1,291
Share of revenue: Approximately 68.13%

Champions are highly valuable customers who demonstrate strong purchasing behaviour.

They are important because they account for the largest share of historical revenue in the revised analysis.

Recommended actions:

- Provide loyalty rewards and exclusive offers.
- Encourage repeat purchases through personalized recommendations.
- Offer early access to new products.
- Encourage reviews and referrals.

7.2 Loyal Customers

Customers: 1,147
Share of revenue: Approximately 14.92%

Loyal Customers have established purchasing relationships with the business.

Recommended actions:

- Introduce loyalty programs and tier-based benefits.
- Recommend complementary products.
- Use personalized offers to increase order value.
- Encourage continued engagement.

7.3 Can't Lose Them

Customers: 224
Share of revenue: Approximately 5.67%

This segment represents customers with valuable historical purchasing behaviour who may have become inactive or are at risk of disengaging.

Recommended actions:

- Prioritize personalized win-back campaigns.
- Offer relevant incentives based on previous purchases.
- Send targeted reminders.
- Monitor whether customers return after a campaign.

7.4 At Risk

Customers: 389
Share of revenue: Approximately 2.98%

At Risk customers may have purchased previously but have not returned recently enough to maintain the same level of engagement.

Recommended actions:

- Launch targeted reactivation campaigns.
- Highlight relevant new products or offers.
- Use email reminders and personalized recommendations.
- Track campaign response and subsequent purchases.

7.5 Potential Loyalists

Customers: 359
Share of revenue: Approximately 2.29%

Potential Loyalists show promising purchase behaviour and may become more valuable with suitable engagement.

Recommended actions:

- Encourage a second or subsequent purchase.
- Recommend products based on past transactions.
- Offer introductory loyalty benefits.
- Test targeted promotions to encourage repeat purchases.

7.6 Needs Attention

Customers: 1,134
Share of revenue: Approximately 4.04%

These customers may need additional engagement to prevent further decline in purchasing activity.

Recommended actions:

- Send relevant product recommendations.
- Use targeted reminders or limited-time promotions.
- Monitor changes in purchase frequency.
- Avoid excessive discounts without measuring their effectiveness.

7.7 Hibernating

Customers: 1,280
Share of revenue: Approximately 1.86%

Hibernating customers have relatively weak recent engagement and limited purchasing activity.

Recommended actions:

- Test low-cost reactivation campaigns.
- Offer relevant products based on purchase history.
- Use a controlled budget for win-back offers.
- Reduce marketing spend on customers who do not respond.

7.8 New Customers

Customers: 57
Share of revenue: Approximately 0.11%

New Customers have recent purchasing activity but limited purchase history.

Recommended actions:

- Send a welcome communication.
- Provide useful product recommendations.
- Encourage a follow-up purchase.
- Introduce the benefits of the loyalty program.

Segment revenue shares are rounded. The segmentation rules and score thresholds are those implemented in the revised notebook.

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8. Customer Reactivation Analysis

A major objective of this project was to identify customers who could be targeted for reactivation.

The revised analysis focused on the **Can't Lose Them** and **At Risk** segments.

Key Reactivation Findings

Metric	Result
Customers identified for reactivation	613
Historical revenue from these customers	£1,502,621.66
Share of total historical revenue	Approximately 8.65%

These customers represent a potential retention opportunity because they previously generated meaningful revenue but may no longer be purchasing regularly.

The historical revenue associated with these customers should not be interpreted as revenue that the business will necessarily recover. Actual recovery depends on campaign effectiveness, customer response, and future purchasing behaviour.

Marketing Priority Analysis

The notebook further divided the reactivation group into three marketing priority levels.

Priority	Customers	Historical revenue
High	96	£694,270.03
Medium	148	£400,235.12
Low	369	£408,116.51
Total	**613**	**£1,502,621.66**

Recommended Marketing Strategy

High priority: Start with personalized outreach to customers with the strongest historical value. Use tailored offers and direct win-back communications.

Medium priority: Test targeted email campaigns, relevant recommendations, and selective incentives.

Low priority: Use cost-effective campaigns, such as automated reminders and broader re-engagement emails. Evaluate responses before investing in expensive offers.

This prioritization helps allocate marketing resources more effectively rather than treating every inactive customer equally.

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9. Power BI Dashboard

Power BI was used to present the project's key metrics and customer-segmentation results in a business-friendly format.

The dashboard is intended to help users understand overall performance, compare customer groups, and identify potential retention opportunities.

Suggested Dashboard Components

Page 1 — Business Overview

- Total Revenue
- Total Orders
- Total Customers
- Total Products
- Average Order Value
- Revenue trend over time
- Revenue by customer segment

Page 2 — RFM Customer Segmentation

- Customer count by segment
- Revenue contribution by segment
- RFM segment distribution
- Segment-level comparison of Recency, Frequency, and Monetary values

Page 3 — Customer and Product Analysis

- Top customers by revenue
- Top products by revenue
- Top products by quantity sold
- Top countries by revenue

Product-level charts should exclude non-product transaction entries such as postage and manual adjustments when those entries are not genuine products.

Dashboard Validation

Before publishing the dashboard, its KPIs and segment counts should be checked against the revised notebook.

The correct revised notebook benchmarks are:

- **5,881 customers**
- **36,975 orders**
- **4,631 products**
- **£17,374,804.27 revenue**
- **£469.91 average order value**

The customer count may differ if Power BI uses a different table, filtering rule, or definition of a unique customer. Confirm the relationship between the sales table and the RFM table, the distinct-count calculation, and any active filters before changing the notebook figures.

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10. Key Business Insights

The revised analysis produced several important findings.

Insight 1: Champions Drive Most Historical Revenue

Champions account for approximately 68.13% of total historical revenue despite representing 1,291 of the 5,881 customers.

Business implication: Protect these customer relationships through retention campaigns, loyalty benefits, and personalized engagement.

Insight 2: Loyal Customers Represent Another Important Revenue Group

Loyal Customers contribute approximately 14.92% of historical revenue.

Business implication: Encourage repeat purchases and strengthen loyalty to support long-term customer value.

Insight 3: Reactivation Is a Meaningful Opportunity

The Can't Lose Them and At Risk segments contain 613 customers associated with approximately £1.50 million in historical revenue.

Business implication: Prioritize targeted win-back campaigns, starting with the high-priority group, and measure the actual revenue recovered.

Insight 4: Many Customers Have Weak Engagement

Hibernating and Needs Attention together account for 2,414 customers.

Business implication: Develop cost-effective engagement strategies and measure whether they increase purchase frequency or bring customers back.

Insight 5: New Customers Need Follow-Up Engagement

The New Customers segment contains 57 customers.

Business implication: Welcome communications and relevant product recommendations may help convert initial purchases into repeat purchasing behaviour.

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11. Project Limitations

The project has several limitations that should be considered when interpreting the findings.

1. **Historical data:** RFM describes past purchasing behaviour and does not guarantee future purchases.
2. **Missing Customer IDs:** Transactions without usable customer identifiers cannot be reliably attributed to individual customers and are excluded from customer-level segmentation.
3. **Scoring methodology:** Quantile-based scoring and the selected segment rules influence how customers are classified.
4. **No campaign experiment:** The project identifies potential marketing opportunities but does not establish that a particular campaign will cause increased revenue.
5. **Observation period:** Customer behaviour is evaluated over the available historical period, so the results may not represent current customer activity.
6. **Revenue interpretation:** Historical revenue associated with a segment is not the same as recoverable revenue or expected future revenue.

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12. Future Improvements

The project could be extended in the following ways:

- Measure customer retention and repeat-purchase rates over time.
- Compare reactivation campaign performance before and after implementation.
- Estimate customer lifetime value.

- Build predictive models for churn or future spending.
- Develop automated customer lists for targeted marketing.
- Add cohort analysis to understand how customer behaviour changes over time.
- Monitor dashboard KPIs as new transaction data becomes available.

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13. Conclusion

This project used Python-based data cleaning, exploratory analysis, and RFM scoring to segment 5,881 identifiable customers from the Online Retail II dataset.

The revised analysis identified eight customer segments and showed that Champions contribute the largest share of historical revenue. It also highlighted a reactivation group of 613 customers associated with approximately £1.50 million in historical revenue.

The findings provide a structured basis for customer-retention decisions, targeted marketing, and resource allocation. The Power BI dashboard complements the Python analysis by presenting the metrics and segments in a visual format.

The project demonstrates practical skills in data cleaning, customer-level aggregation, feature engineering, quantile-based scoring, business analysis, and dashboard reporting.

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14. Tools and Skills Demonstrated

Python

- Pandas
- NumPy
- Data cleaning and transformation
- Grouping and aggregation
- Customer-level feature engineering

Analytics

- Exploratory Data Analysis
- RFM methodology
- Customer segmentation
- Revenue analysis

- Marketing prioritization
- Business insight generation

Power BI

- KPI reporting
- Customer-segment visualization
- Revenue analysis
- Interactive dashboard development

Business Skills

- Translating analytical results into business recommendations
- Identifying customer-retention opportunities
- Prioritizing marketing actions using customer value
- Communicating data-driven insights